Practical Organic Chemistry

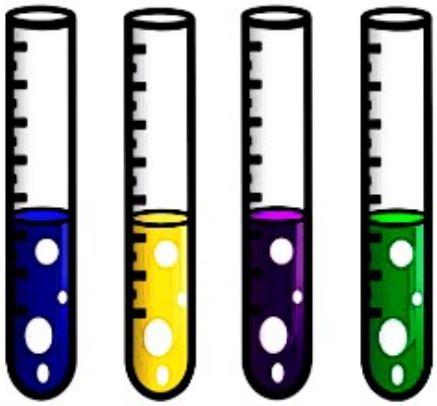
DISCLAIMER

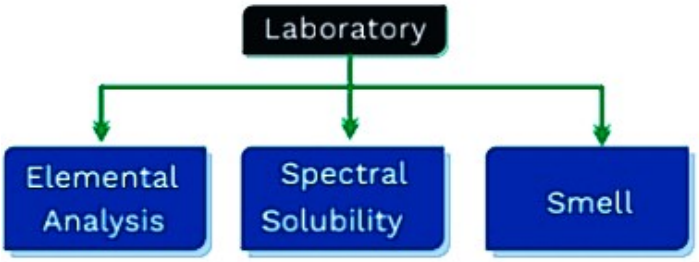
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Practical Organic Chemistry

INTRODUCTION

Why we required a test?
What kind of reasonable observation is suitable for particular test.
Major component or any test in laboratory
We need to characterize any reaction product or 2 or more than two element / compound/ion.
Practical organic chemistry give us exact idea about desired information.





About Alcohol

(a) Colourless liquid (At 25°C)
(b) Blower alcohol are free flowing, fruity smell and liquid.
(c) Branched (highly) and more than 12 carbon are solids.

Test For Alcohols :

(i) Sodium Metal

When Na-metal reacts with alcohol, $H_2(g)$ evolve

$$\underset{\text{Alcohol}}{ROH} + Na \longrightarrow \underset{\text{(Sodium alkoxide)}}{RO^-Na^+} + \frac{1}{2}H_2(g)$$

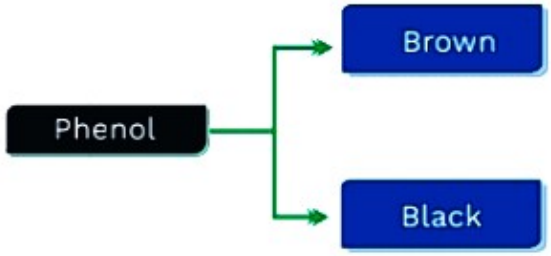
(ii) Ceric Ammonium Nitrate Oxidation (CAN Test)

$$\underset{\text{ceric ammonium nitrate (yellow)}}{(NH_4)_2Ce(NO_3)_6} + \underset{\text{alcohol}}{ROH} \longrightarrow \underset{\text{(Red)}}{(NH_4)_2Ce(OR)(NO_3)_5} + NHO_3 \longrightarrow \underset{\text{colour decolourise}}{Ce(III)}$$

Points to remember

- Sodium metal test is possible in 1°, 2° & 3°
- The rate is highly variable and depends upon the alcohol structure.
- Other Functional groups that evolve $H_2(\uparrow)$ R_2NH, RSH, $RC{\equiv}C{-}H$, RCO_2H etc.

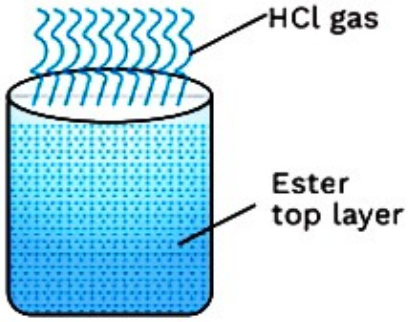
- Use :
- Identification test for 1° and 2°, but in case 3° alcohol it is very slow.
- Note :



Points to remember

- Colour yellow →Red → Colourless [1 min to 12 hrs.]
- Oxidation with Tones Reagent (CrO_3)

(iii) Jones Oxidation: Chromic anhydride or Chromium Trioxide (CrO_3)



Points to remember

- 1° and 2° alcohol → Intense Blue to Green
- Aldehyde → blue to green

(iv) Lucas Test - This test distinguishes 1°, 2° & 3° alcohol

- It uses, anhydrous $ZnCl_2$ in conc. HCl (lucas reagent)

$$R-OH \xrightarrow{(HCl+ZnCl_2)} R-Cl + H_2O$$

(v) Victor Mayer Test: This is also used to distinguishes 1°, 2° & 3° alcohol

Points to remember

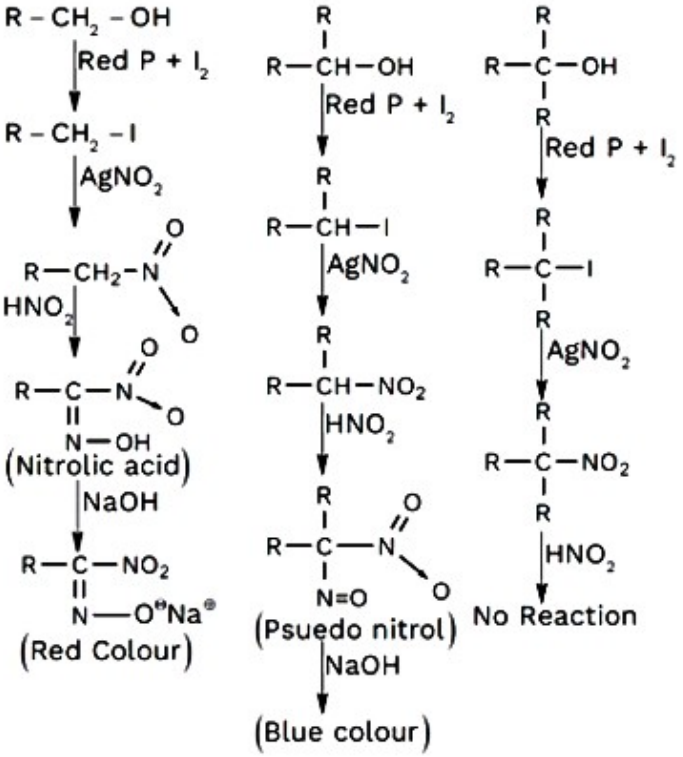
This test (Lucas Test) distinguishes 1°, 2° & 3° alcohol from each other.

3° alcohols give white turbid solution immediately

2° alcohols give white turbid solution within 5 to 10 minutes.

1° alcohols do not give white turbidity at normal temperature.

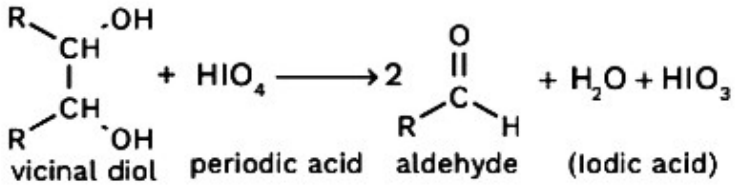
1°-Alcohol	2°-Alcohol	3°-Alcohol



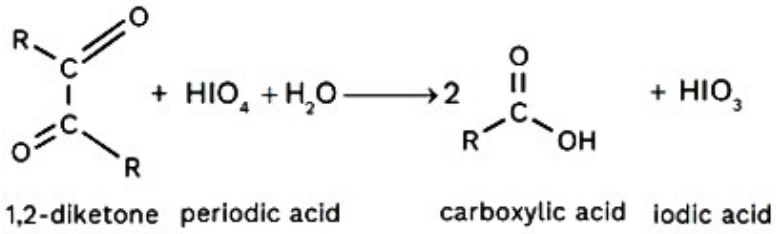
(vi) Periodic Acid (Reagent = 5% $AgNO_3$ + HIO_3)

***Test for detection of Vicinal Diols and Related Compounds**

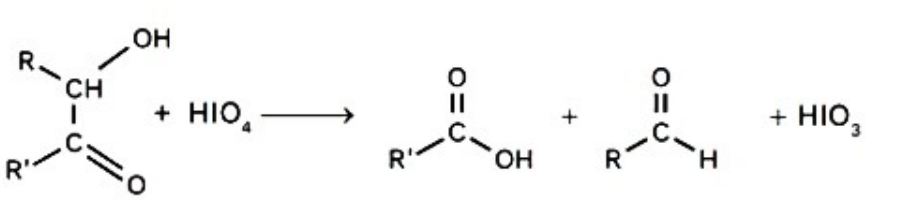
1.



2.

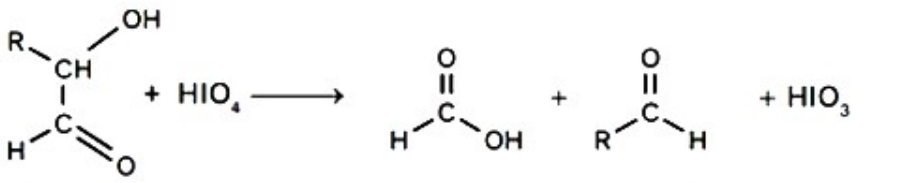


3.



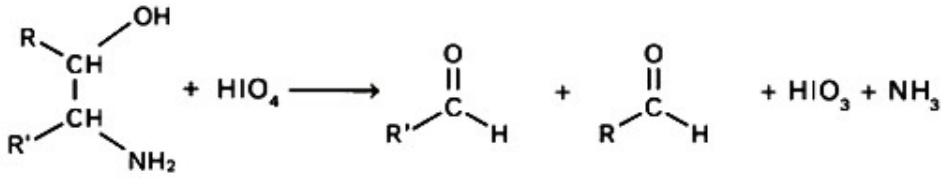
α –hydroxy ketone | periodic acid | carboxylic acid | aldehyde | iodic acid

4.



α –hydroxy aldehyde | periodic acid | carboxylic acid | aldehyde | iodic acid

5.



β –hydroxy amine | periodic acid | aldehyde | iodic acid

Reaction envolved:

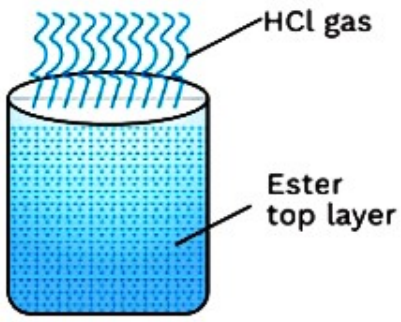
$$\underset{\text{Iodic acid}}{HIO_3} + \underset{\text{Silver nitrate}}{AgNO_3} \longrightarrow HNO_3 + \underset{\text{Silver iodate (white)}}{AgIO_3(s)}$$

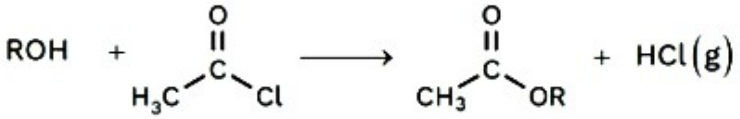
Points to remember

No effect on functional group olefins, 2° alcohols, 1, 3-glycols, Ketone and Aldehydes.

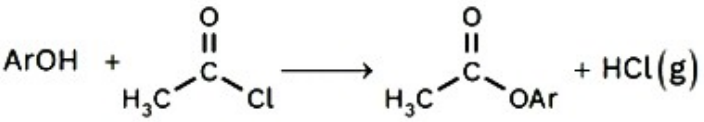
(vii) Acetyl Chloride

Observation:





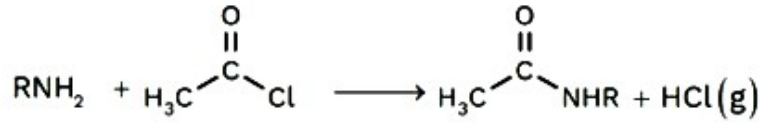
alcohol acetyl chloride ester



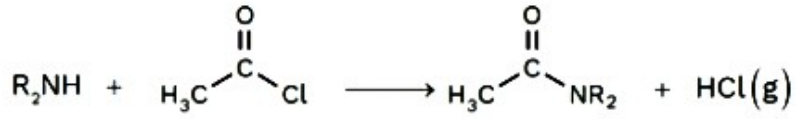
phenol acetylchloride ester

NOTE: 3° alcohols form primarily alkyl chloride due to the reaction of the liberated HCl on another molecule of the alcohol.

- Other functional groups that give positive test: 1° and 2° amines



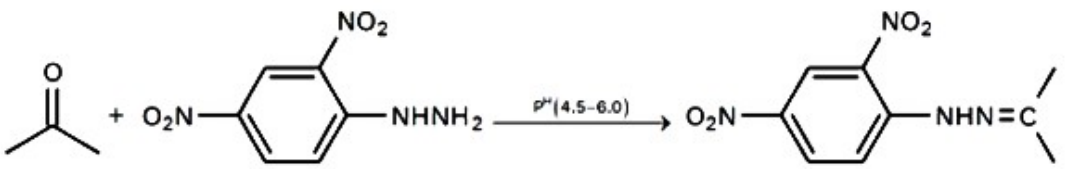
1° amine acetyl chloride amide



2° amine acetyl chloride amide

2. Classification Tests for Aldehydes and Ketone

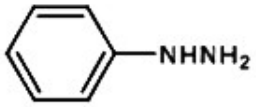
(i) 2,4-dinitrophenyl Hydrazine



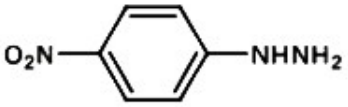
aldehyde or ketone 2,4-dinitrophenylhydrazine 2,4-dinitrophenylhydrazone

Observation: Formation of yellow, orange or red ppt. initial appearance will be oily, but after same time it is crystalline.

Phenyl hydrazine and p-Nitrophenyl hydrazine.



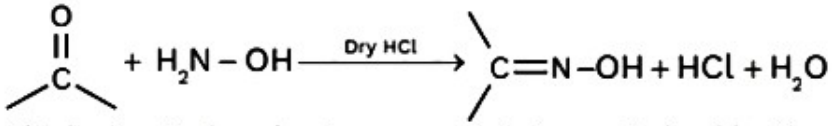
phenylhydrazine



p-nitrophenylhydrazine

Positive Test: formation of yellow ppt.

(ii) Hydroxylamine :



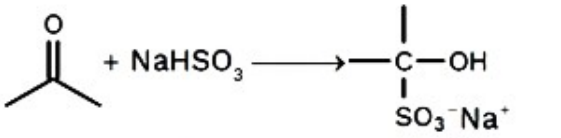
Aldehyde or ketone | Hydroxylamine | Oxime | Hydrochloride

Observation



H_2S gas $\xrightarrow[\text{test}]{PH}$ Orange $\rightarrow$ Red

(iii) Sodium Bisulphite ($NaHSO_3$)



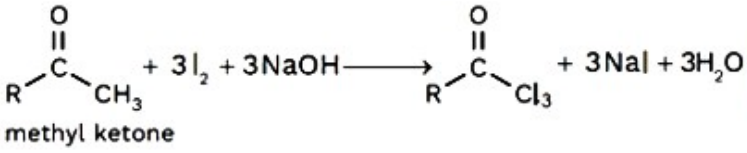
Aldehyde or ketone | Sodium bisulphite | White crystalline adduct

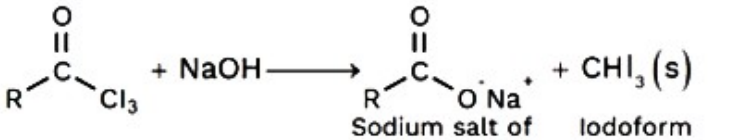
Points to remember

Positive Test: By aldehydes and methyl ketone

- Only some ketone (At least one-CH_3 group) and aldehyde.
- Even cyclic ketone also feel difficulty in test.

(iv) Iodoform Test (For Methyl Ketones)

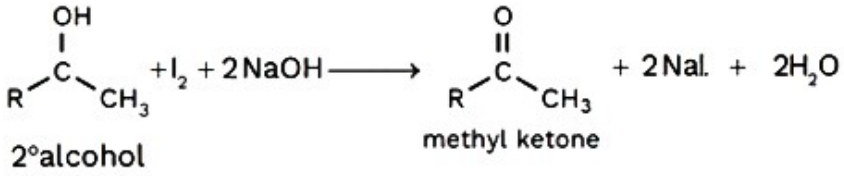




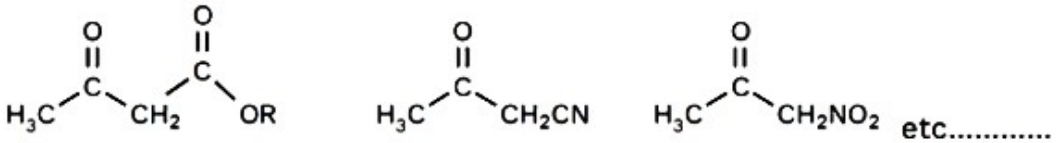
Points to remember

- **Observation :** Yellow ppt.
- **Draw back:** 2° alkyl halide also show positive test due to oxidation.

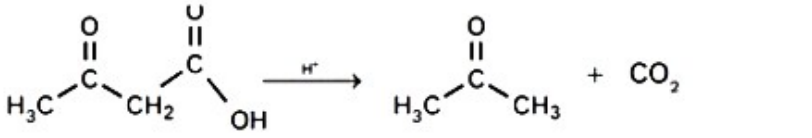
- The principal type of compound that give a positive test:

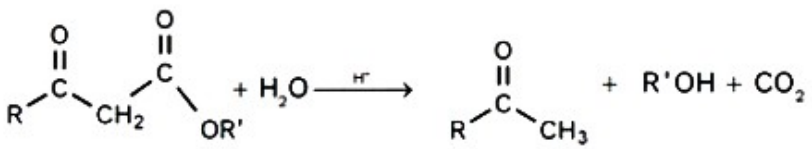


- The test is negative for the following type:



- Reaction favour decarboxylation instead of halofrom reaction:





After decarboxylation, methyl carbonyl group presence will favour iodofrom test

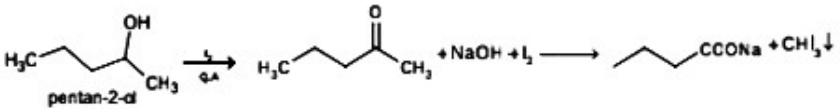
Q1 **A compound that gives a positive iodoform test is:**

(A) 1-Pentanol **(B) 3-Pentanol**

(C) 2- Pentanol **(D) Pentanol**

A1 **C**

For iodoform test, $CH_3-\overset{O}{\overset{\|}{C}}-$ group is required

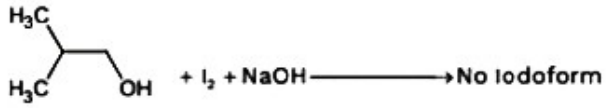


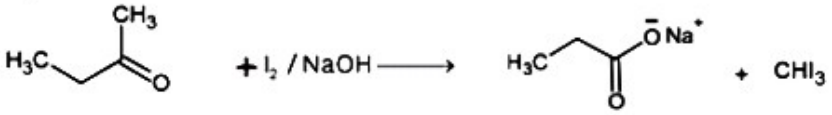
Q2 **Iodoform can be prepared from all except.**

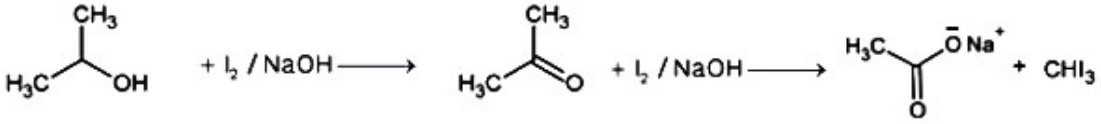
(A) Isobutyl alcohol **(B) Ethyl methyl ketone**

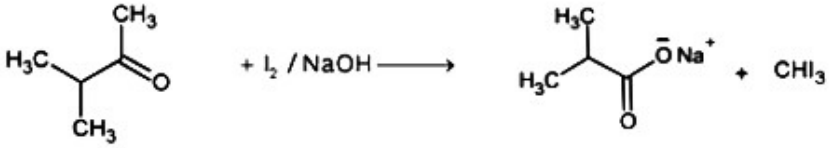
(C) Isopropyl alcohol **(D) 3-Methyl-2-butanone**

A2 **A**

(A) 

(B) 

(C) 

(D) 

Q3 **Which of the following cannot be used for the preparation of iodoform ?**

(A) Acetone **(B) Methanol**

(C) Ethanol **(D) Acetaldehyde**

A3 **B**

$$CH_3 - OH \xrightarrow[NaCH]{I_2} \text{No iodoform}$$

3. Tests that give positive results with aldehydes and negative results with Ketones

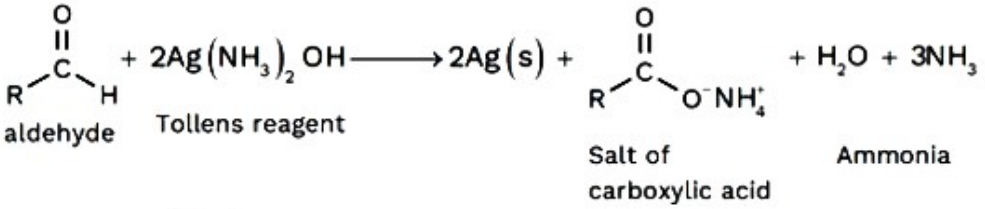
Introduction:

(i) Carbonyl polar due to C=O bond.
(ii) Soluble in water considerable hydrogen bond acceptors.
(iii) Carbonyl oxygen in weakly basic.
(iv) Oxidation is possible in aldehyde and ketone both.

(i) Jones Reagent (CrO_3) chromium trioxide

(ii) Tollens Reagent $Ag(NH_3)_2OH$

Tollens' reagent is usually used as a test for aldehydes, with which it gives a characteristic silver mirror:



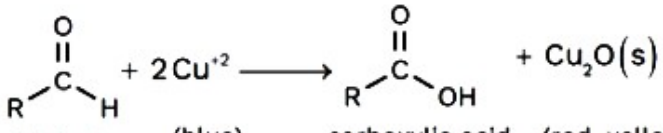
aldehyde — Tollens reagent — Salt of carboxylic acid — Ammonia



Points to remember

Positive Test: Silver mirror formation in test tube or silver ppt formation

(iii) Bendicts solution and Fehling's Solution

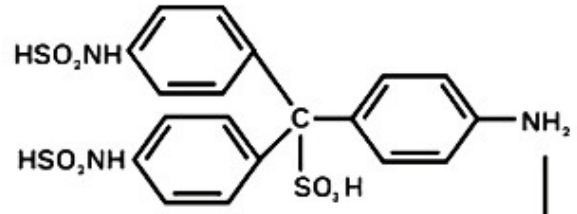


aldehyde — (blue) — carboxylic acid — (red, yellow, or yellowish green)

Points to remember

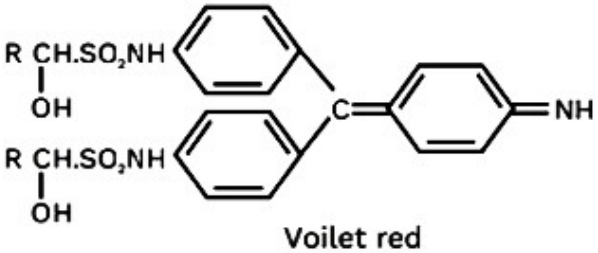
- Yellow or Yellowish Green.
- Aromatic aldehyde do not give positive result.

(4) Schiff's reagent



Schiff's reagent colourless

$2R.CHO$
$-H_2SO_4$



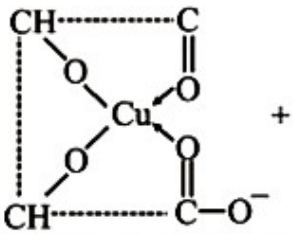
Voilet red

Let's understand the procedure

2 ml of substance, if it is liquid or to its 2 mol cold aqueous solution, add 2ml Schiff's reagent.

(Rosaniline decolourised with SO_2) and shake for 2 minuts. Do not heat or warm. A deep violet-red or red colour indicates the presence of aldehydic group.

(5) Fehling's test –.



$+R.CHO+2H_2O$

Tartrato-cuprate anoin (blue)
(Fehling's solution)

$\xrightarrow{\Delta} 2\left|\begin{matrix}CHOH.COO^- \\ CHOH.COO^-\end{matrix}\right. + R.COOH + Cu_2O\downarrow$

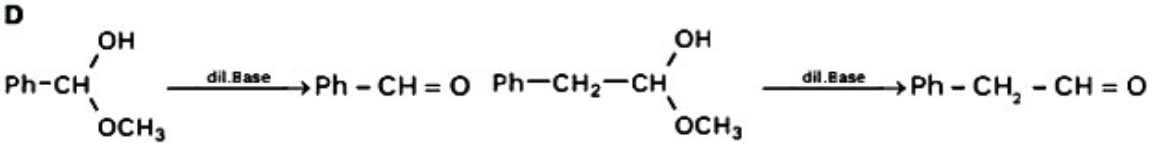
cuprous oxide (Red)

Let's understand the procedure

In a test tube mix 1 ml each of Fehling A ($CuSO_4$ in water with few drops of acetic acid) and Fehling B (Rochelle salt and NaOH solution) solutions when a deep blue colour appears. To this add 0.1gm or 2-3 drops of substance and boil for 5 minuts. The blue colour of Fehling's solution slowly fades away and a reddish-brown precipitate of Cu_2O is obtained indicating the presence of aldehydic group.

Q4 **Which of the following can reduce Fehling solution ?**

A4 **D**



Aromatic aldehydes do not give Fehling solution test.

Q5 **When acetaldehyde is treated with Fehling's solution, it gives a precipitate of:**

(A) Cu **(B) CuO**

(C) Cu_2O **(D) Cu + Cu_2O + CuO**

A5 **C**

$$CH_3 - CHO + \text{Fehling solution} \longrightarrow \underset{\text{red}}{Cu_2O}\downarrow + CH_3COO$$

Q6 **The reagent with which both acetaldehyde and acetone react easily is:**

(A) Tollen's reagent **(B) Schiff's reagent**

(C) Grignard reagent **(D) Fehling reagent**

A6 **C**

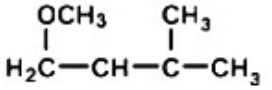
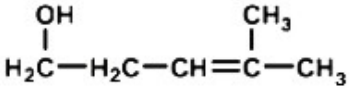
Grignard's reagent reacts with both aldehydes and ketones while other three reagents react only with aldehydes, not with ketons.

Q7 **The reagent which will be suitable to distinguish 1-methoxy-3-methylbut-2-ene from isomeric 4-methyl-3-pentene-1-ol is:**

(A) Bromine in chloroform **(B) Alkaline potassium permanganate**

(C) Ammoniacal silver nitrate **(D) Sodium metal**

A7 **D**

Lab test		
$Br_2/CHCl_3$	+ ve	+ ve
Alkaline $KMnO_4$	+ ve	+ ve
$[Ag(NH_3)_2]\ NO_3$	- ve	- ve
Na metal	- ve	+ ve

Q8 **Three isomeric compounds M,N and P ($C_5H_{10}O$) give the following tests:**

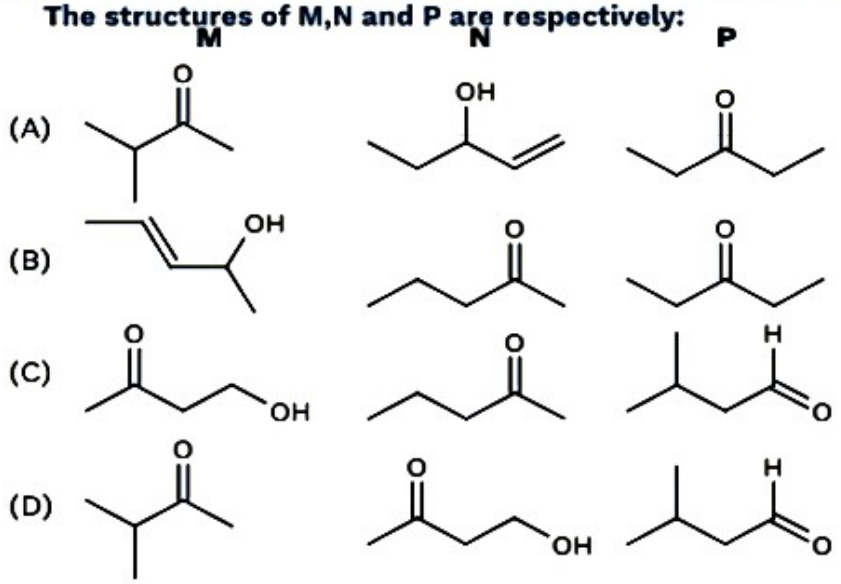
(i) M and P react with sodium bisulphite to from an adduct.

(ii) N consumes 1 mol of bromine and also gives turbidity with conc. HCl/anhydrous $ZnCl_2$ after prolong heating.

(iii) M reacts with excess of iodine in alkaline solution to give yellow crystalline compound.

(iv) p-Rosaniline treated with sulphur dioxide develops pink colour on shaking with P.

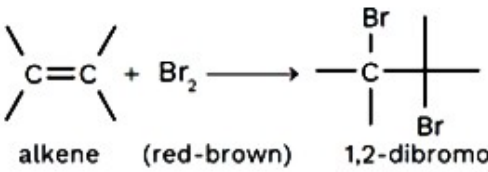
The structures of M,N and P are respectively:



A8 **D**

4. Classification Tests for Unsaturation "alkenes & alkynes"

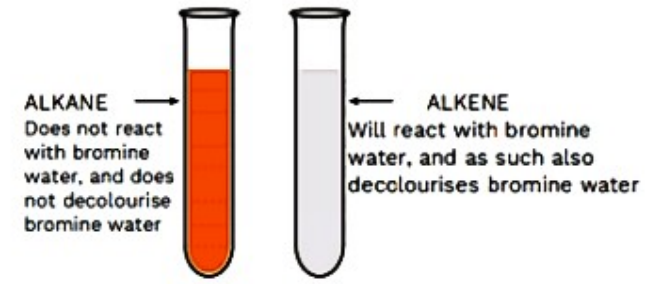
(i) CCl_4/Br_2:



alkene (red-brown) 1,2-dibromoalkane (colourless)

Do you know?

- Alkanes are not usually characterized chemically.
- Inert behavior of alkene do not attract electrophile easily.



ADDITION OF BROMINE WATER TO ALKANES & ALKENES

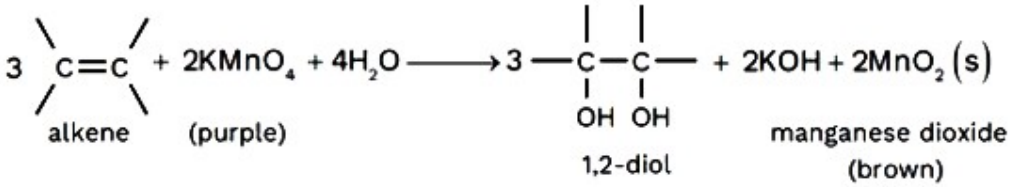
Points to remember

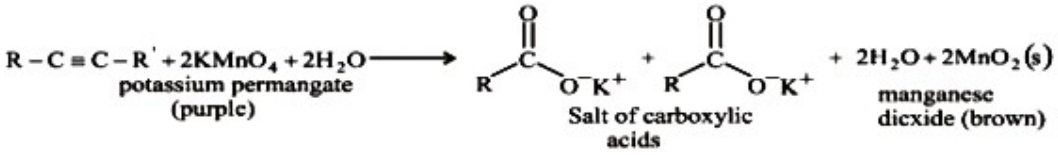
(a) Colour discharge without edition of HBr

(b)  and  Positive test

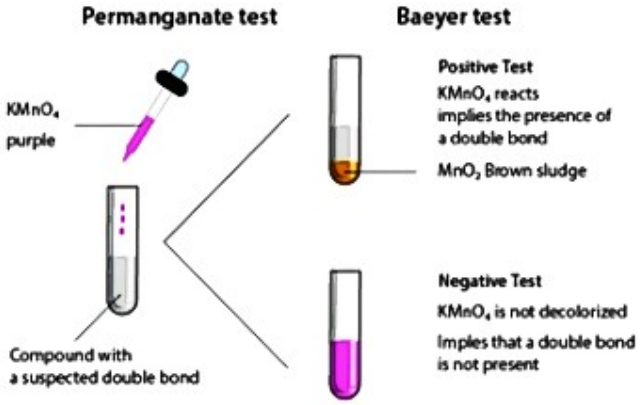
(c) Br_2/CCl_4 + Amine → Salt + decolouration

(ii) Baeyer Test ($KMnO_4$ aqueous)





Pure colour disapplears, and brown colour ppt. (MnO_2) appears.



Reaction -1 $C_6H_5CH = CH_2 + Br_2 \xrightarrow{\text{rapid}} C_6H_5\overset{Br}{\overset{|}{C}}HCH_2Br$

Reaction - 2 $C_6H_5CH = CHCOOH + Br_2 \xrightarrow{\text{slow}} C_6H_5\overset{Br}{\overset{|}{C}}H\overset{Br}{\overset{|}{C}}HCOOH$

Observation: Reaction-1 and Reaction-2 both show addition reaction (molecules are unsaturated), Still reaction -1 show higher rate of reaction in case of Br_2 /CCl_4 , while in case of $KMNO_4$ reaction-2 will be fast due to electron withdrawing group attached with double bond. It show some complementary realations in both reactions.

Points to remember

- $H_5C_6-\overset{Br}{\overset{|}{C}}=\overset{Br}{\overset{|}{C}}-C_6H_5$ and $(Ph)_2C=C(Ph)_2$ like tetrasubstituted olefins do not slow positive test.
- Br_2/CCl_4 or KMnO4 both reagent fail in case of sterically hindered indeeules.

Q9 **Which of the following will decolorize alkaline $KMnO_4$ solution?**

(A) C_3H_8 **(B) C_6H_6(Benzene)**

(C) CCl_4 **(D) C_2H_4**

A9 **D**

Alkene/Alkyne decolourises the pink/purple colour of alkaline $KMnO_4$ solution

Q10 **Which of the following compound does not dissolve in conc. H_2SO_4 even on warning?**

(A) Ethylene **(B) Benzene**

(C) Hexane **(D) Aniline**

A10 **C**

Ethylene absorbs H_2SO_4 forming $CH_3 - CH_2OSO_3H$ and dissolve.
Benzene with warm H_2SO_4 undergoes sulphonation, and dissolve.
Aniline with H_2SO_4 forms anilinium hydrogen sulphate salt and dissolve.
Hexane, a hydrophobic molecule, does not react with H_2SO_4, remains insoluble.

Q11 **What simple laboratory test could be performed to distinguish between 1-Pentyne and 2-Pentyne?**

(A) The addition of Ag^+ in ammonia

(B) The addition of H_2SO_4 in Hg^+

(C) The addition of Br_2 in CCl_4

(D) The addition of H_2 on a Pd catalyst with $CaCO_3$

A11 **A**

Terminal alkyne reacts with silver ion and form white precipitate.

Points to remember

- allylic and benzylic RX give positive result
- The identity of the halogen can sometimes be determined from the colour of the silver halide: AgCl (white); AgBr (pale yellow); AgI (yellow)

5. Test for halogen (alkyl) derivatives :

Primary test is qualitative halogen analysis.

(i) Ethanolic silver nitrate

(ii) Sodium iodide in acetone

Carbocation (S_N1) formation is R.D.S

$AgNO_3$/ethenol or NaI/Acetone proceeds by a direct displacement

$AgNO_3$/ethanol test: $R_3CX > R_2CHX > RCH_2X$

NaI/acetone test: $R_3CX < R_2CHX < RCH_2X$

Points to remember

$\boxed{2° \text{ and } 3° R-X}$ → Positive

R – X = Alkyl halide

$\boxed{1° \text{ R} - \text{X, Ar} = \text{X and Vinyl halide}}$ → Negative test

$\boxed{\text{Allylic and benzylic alkyl halide}}$ → Positive

(A) EtOH/$AgNO_3$

$$\underset{\text{alkyl halide}}{RX} + \underset{\text{silver nitrate}}{AgNO_3} \xrightarrow{CH_3CH_2OH} \underset{\text{silver halice}}{AgX(s)} + RONO_2$$

(B) Sodium Iodide in Acetone Test

Uses:

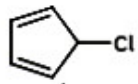
(i) To identify Br or I element

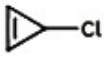
(ii) Precipitation from (NaX) is key concept.

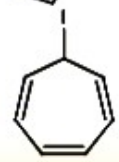
$$\underset{\text{Alkyl chloride}}{RCl} + \underset{\text{Sodium iodide}}{NaI} \xrightarrow{\text{Acetone}} \underset{\text{Alkyl iodide}}{RI} + \underset{\text{Sodium chloride}}{NaCl(s)}$$

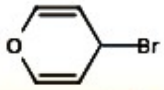
$$\underset{\text{Alkyl bromide}}{RBr} + \underset{\text{Sodium iodide}}{NaI} \xrightarrow{\text{Acetone}} \underset{\text{Alkyl iodide}}{RI} + \underset{\text{Sodium bromide}}{NaBr(s)}$$

Q12 **Identify the compound which can give white precipitate with aq. $AgNO_3$:**

(A) 

(B) 

(C) 

(D) 

A12 **B**

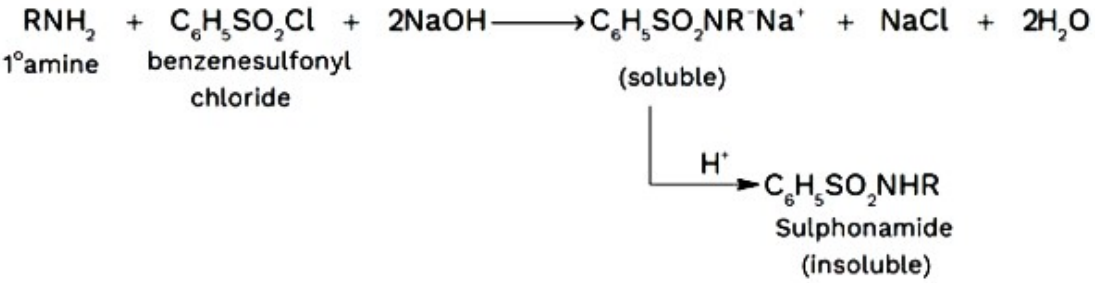
In 'a' carbocation will be anti-aromatic so, does not respond with $AgNO_3$ while 'c & d' react with $AgNO_3$ but colour of precipitate is not white.

6. Tests for Amines and Amine Salts

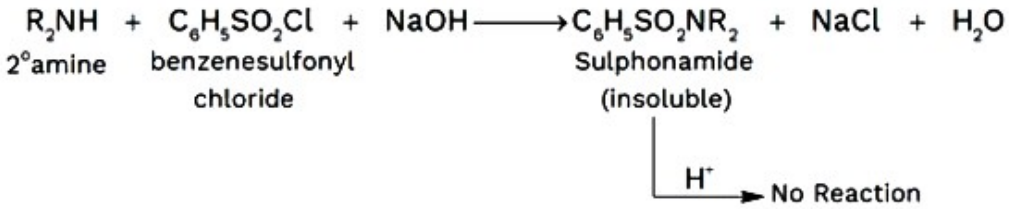
(i) Hinsberg Test

Based on the reaction of the amine with benzene sulfonyl chloride (Heinsberg reagent), it can be used to separate 1°, 2°, and 3° amines.

1° amines: give solution that produce ppt after addition of HCl



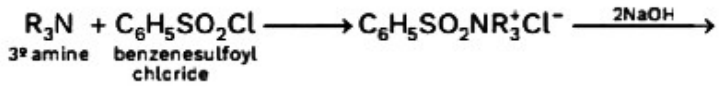
2° amines: give ppt - acidification of the solution does not dissolve the sulphonamide (i.e., the PPT is not soluble in NaOH or HCl)

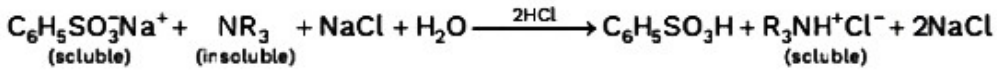


3° amines:

Undergo reaction with benzenesulfonyl chloride to produce quaternary ammonium sulfonate salts, which yield sodium sulfonate and insoluble 3° amines in basic solution.

Acidification of the reaction mixture results in the formation of sulfonic acids and soluble amine salts





Points to remember

- The results of the Hinsberg test must not be used alone in classifying amines. The solubility of the original compounds must also be considered.
- If the original compound is amphoteric, which means that it is soluble in both acids and alkalies, the Hinsberg method fails to distinguish among the types of amines.

Q13 **The compound which reacts with aqueous nitrous acid at low temperature to product an oily nitroso amine is:**

(A) Ammonia **(B) Methylamine**

(C) Dimethylamine **(D) Trimethylamine**

A13 **C**

Secondary amine gives yellow oily nitrosamine on reaction with nitrous acid.

Q14 **A positive biuret test confirms the presence of:**

(A) Ester **(B) Carboxylic acid**

(C) Amide **(D) Amine**

A14 **C**

Q15 **Which of the following reagents on treatment with aniline in basic medium products phenyl isocyanide?**

(A) CCl_4 **(B) Trichloromethane**

(C) Methylene chloride **(D) Hexachloroethane**

A15 **B**

$$\underset{\text{Aniline}}{PhNH_2} + \underset{\text{Trichloro methane}}{CHCl_3} + 3NaOH \xrightarrow[\text{reaction}]{\text{Carbylamine}} \underset{\text{Phenylisocyanide}}{Ph - \overset{\oplus}{N} \equiv \overset{\ominus}{C}} + 3NaCl + 3H_2O$$

Q16 **Hinsberg's reagent is:**

(A) C_6H_5COCl **(B)** CH_3COCl

(C) $C_6H_5CH_2Cl$ **(D)** $C_6H_5SO_2Cl$

A16 **D**

Benzene sulphonyl chloride ($PhSO_2Cl$) is called Hinsber'g reagent. It is used to distinguish and separate the 1°, 2° and 3° amines.

Q17 **Primary, secondary, tertiary amines can be separated by the following except :**

(A) Fractional distillation **(B) Fractional method using diethyl oxalate**

(C) Hinsberg's method using $C_6H_5SO_2Cl$ **(D) Selective crystallization**

A17 **D**

Q18 **1°, 2°, and 3° amines can best distinguished by:**

(A) HNO_2 **treatment** **(B) Exhaustive alkylation**

(C) Mustard oil reaction **(D) Carbylamine reaction**

A18 **A**

Q19 **Which of the following method is neither meant for the synthesis nor for separation of amine?**

(A) Hinsberg method **(B) Hofmann method**

(C) Wurtz reaction **(D) Curtius reaction**

A19 **C**

Wurtz reaction is reaction of alkylhalide with metal to form alkane

$$R - X \xrightarrow[\text{Ether}]{Na} R - R$$

(ii) Libermann's nitroso test

$(CH_3)_2NH + HONO \rightarrow (CH_3)_2N-N = O + H_2O$

Nitrosoamines are water soluble yellow oils and when warmed with phenol and few drops of conc. H_2SO_4 produce a green colour solution which turns blue on adding alkali. This reaction is called Libermann's nitroso reaction .
Tertiary amine do not react with nitrous acid.

Introduction

It is used as a test for secondary amines. Secondary amines (aliphatic as well as aromatic) react with nitrous acid to form N-nitrosoamines.

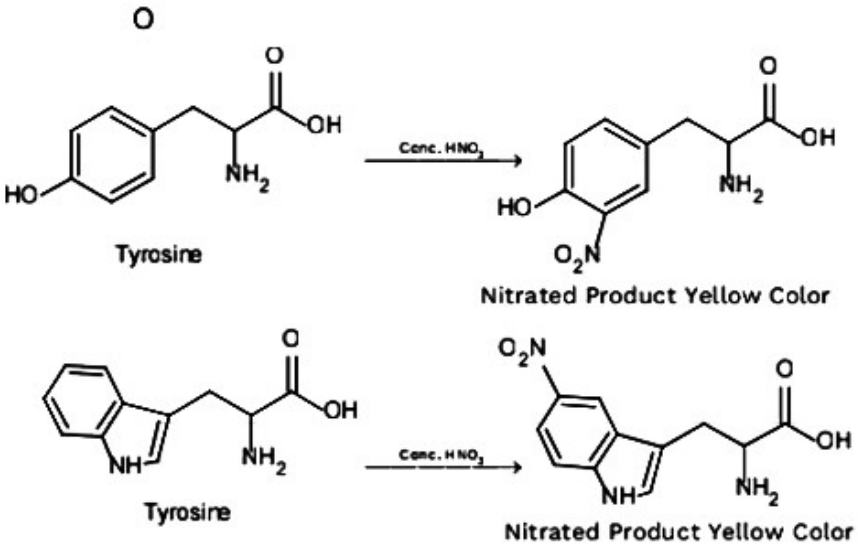
(v) Sodium Metal :
The active hydrogen on 1° and 2° amines undergo reaction with sodium to form salt and liberate hydrogen gas

$$\underset{1^\circ \text{amine}}{2RNH_2} + 2Na \longrightarrow 2RNH^-Na^+ + H_2(g)$$

$$\underset{2^\circ \text{amine}}{2R_2NH} + 2Na \longrightarrow 2R_2N^-Na^+ + H_2(g)$$

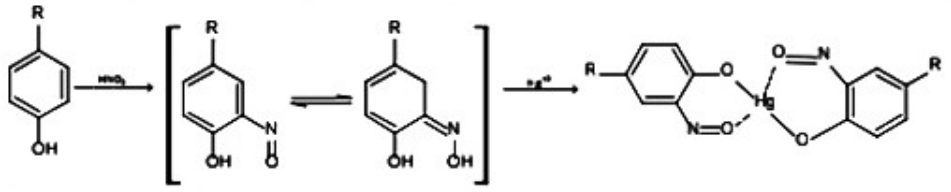
7. Tests for Amino Acids
Test for Protein:

1. Xanthoproteic test: This test is given by protein-containing Tyrosine or Tryptophan



MILLON'S TEST :

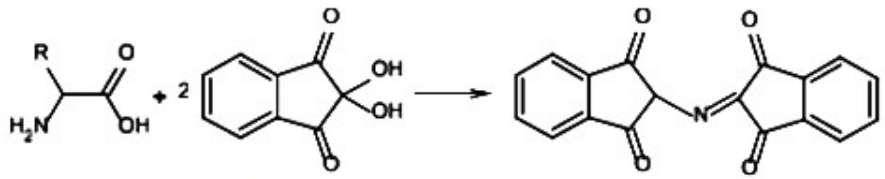
Millin's reagent is a mixture of mercurous and MERCURITE nitrate proteins containing tyrosine given white ppt turning red when treated with Millon's reagent



Ninhydrin Test :

The ninhydrin test is a test for amino acids and proteins with free -NH_2 groups. Amino acids are detected by the ninhydrin test. All amino acid gives violet coloured product with ninhydrin (TRIKETO hydroindene hydrate) except proline and 4hydroxyproline which gives yellow colour with it.

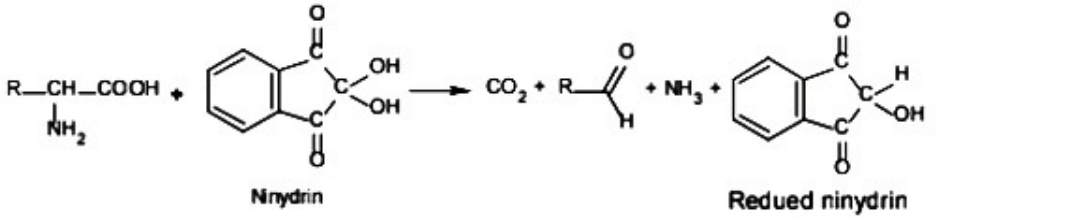
When such as -NH_2 groups reacts with ninhydrin, a purple blue complex is formed



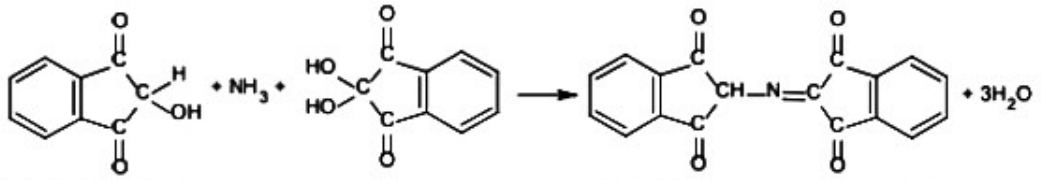
Ninhydrin compound (Purple blue)

The same violet coloured due formed from all α-AAs with 1° amino groups because only their nitrogen is incorporated into it. The 2° amino proline and 4-hydroxyproline give different adducts that absorb light at different and thus hence a different yellow colour.

Rhumann purple ⟶ **Positive Ninthydrin Test**



Ninydrin Redued ninydrin



Redued ninydrin Ninhydrin Violet-blue condensation prduct

Test of Carbohydrates.

Test of Carbohydrates.

(a) Carbohydrate with Tollen's reagent (ammonical silver nitrate) give silver mirror test.

(b) Carbohydrate with Fehling solution (alkaline $CuSO_4$) give a red ppt.

Biological Importance of Carbohydrate:

(i) it is main source of energy.

(ii) it is called fuel of the body.

(iii) Cell wall of plant cell is made up of cellulose.

(iv) Exoskeleton of insect is formed of chitin.

(v) Sugar is structural component of DNA and RNA.

(vi) 1 gm Carbohydrate gives 4.1 kcal of energy.

Molisch's test: Carbohydrates when treated with Molisch's reagent (1% alcoholic solution of α-naphthaline , presence of conc. H_2SO_4 form violet ring.

Molisch's test a general test for all carbohydrates. In this test, carbohydrates when reacted with conc. H_2SO_4 get dehydrated to form furfural and its derivative.

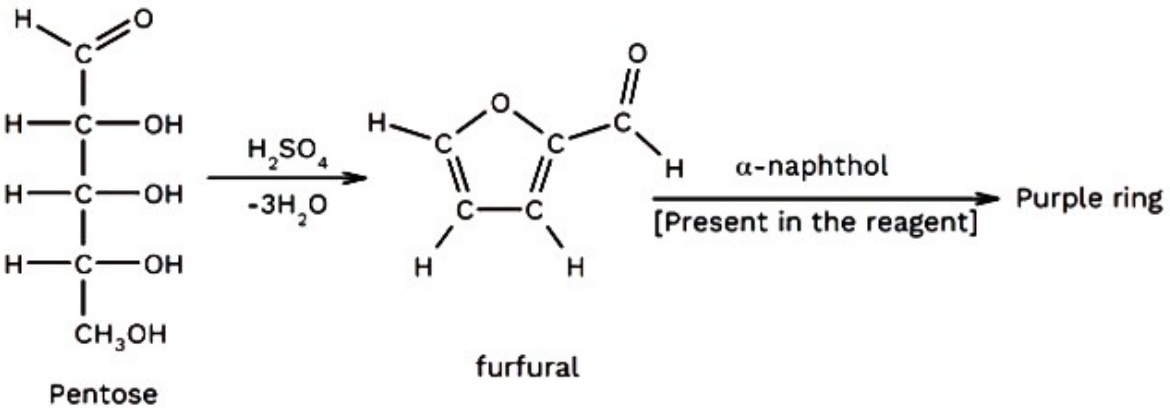
Reagents for Molisch's test:

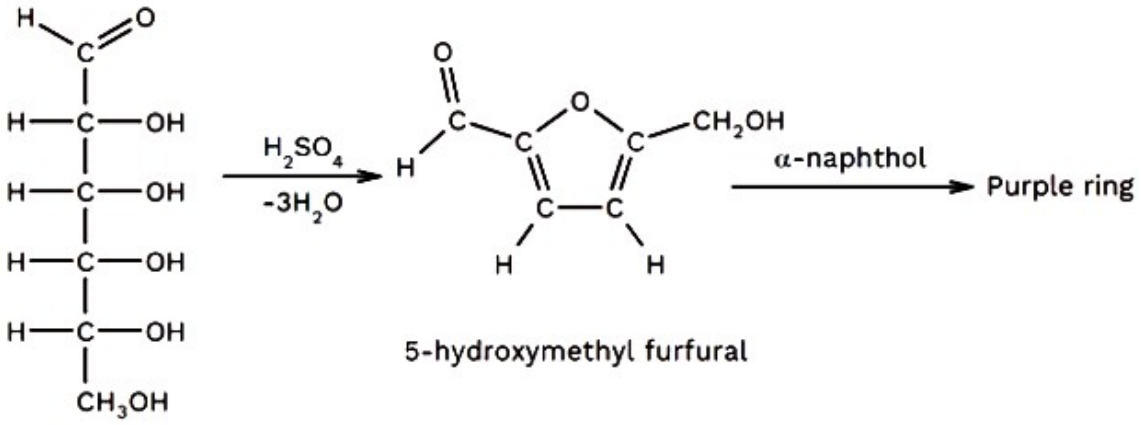
Test solution: 5% Glucose, 5% Sucrose, 5% Starch

Molisch's reagent (5% α-naphthol in ethanol)

H_2SO_4

Pipettes





Hexose

5-hydroxymethyl furfural

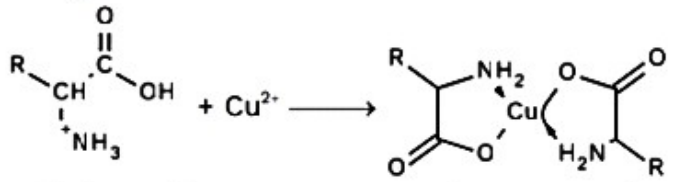
(d) When heated in dry test tube, it melts, turns brown and finally black giving a smell of burning sugar.

Barfoed's Test → **Barfoed's Test is also called Barcode Test**

Shows Positive test for Reducing monosaccharides Reactions:
Reducing monosaccharides are oxidized by the copper ions in solution to form a carboxylic acid and a reddish precipitate of copper (I) oxide with three minutes reducing disaccharides undergo the same reaction, but do so at a shown rate.

How to performs the test:

One ml of a sample solution is placed in a test tube. Three ml of Barford's reagent (a solution of cupric acetate and acetic acid) is added. The solution is then heated in a boiling water both for three minutes.

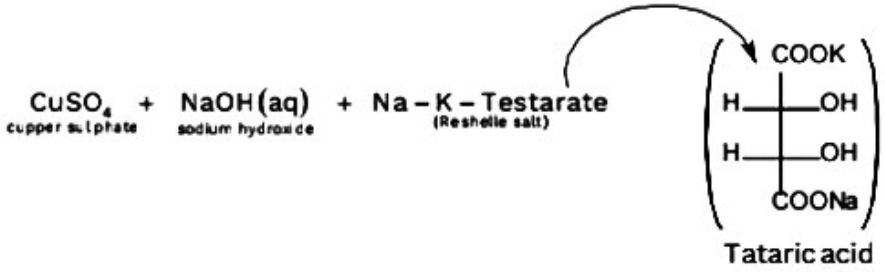


α-Amino acid

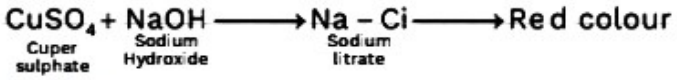
Copper complex
(Blue)

A positive test is indicated by:
The formation of a reddish precipitate within in three minutes

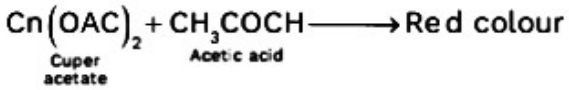
Fehling solution:



Benedict's solution:



Barford Test



But what the difference between there

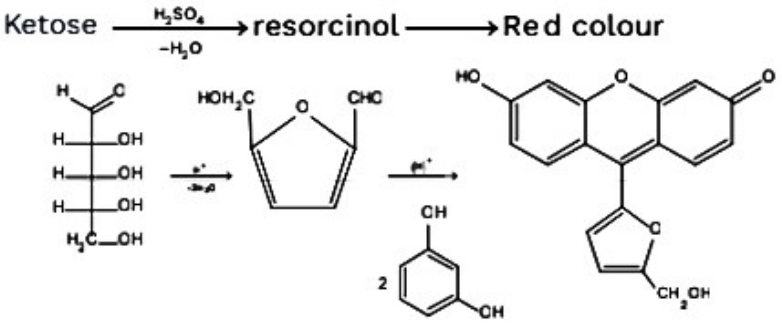
Note:

- Barford Test applicable only monosaccharides
- Fehling and Benedict's Applicable for all monosaccharides, oligosaccharides and polysaccharides

SELIWANOFF'S TEST:

Is a chemical test that distinguishes, between aldose and ketose sugar. Ketoses are distinguished from an aldose Via their ketone/aldehyde functionality.

Seliwinoff only applicable ketose

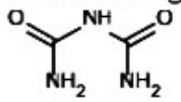


If the sugar contains ketone groups it is a ketose, if a sugar contains an aldehyde group, it is an aldose. When added to a solution containing ketoses, red color is formed rapidly. Indicting a positive test when added to a solution containing aldose, a slower forming light pink is observed instead

. The Biuret test, also known as the Piotrowski test is a chemical test used for detecting the presence of peptides.

BIURET TEST :

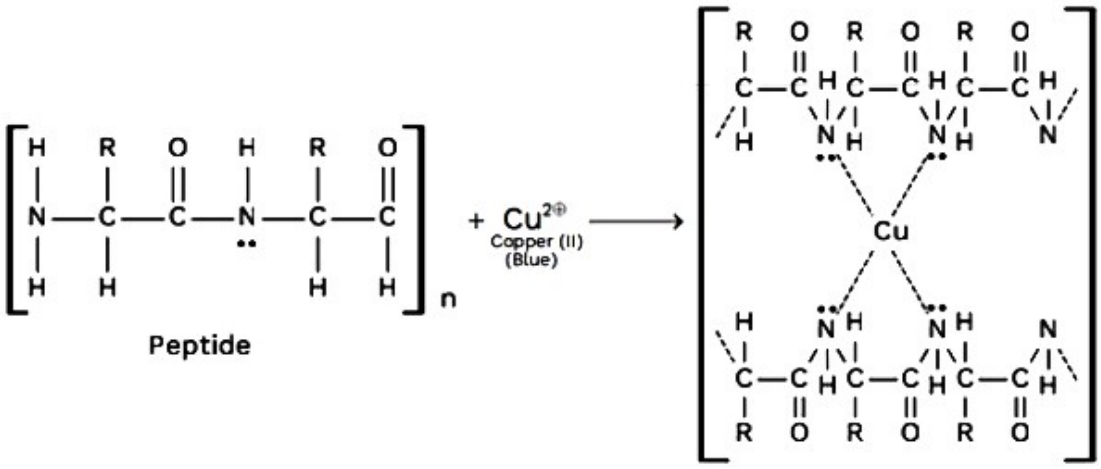
The addition of a very dilute solution of CuSO4 to an alkaline solution of a protein is done. A positive test is indicated by the formation of a pink-violet to purple-violet colour. The name of the tests is derived from specific compounds. Biuret gives positive tests with this regent.



(Biuret)

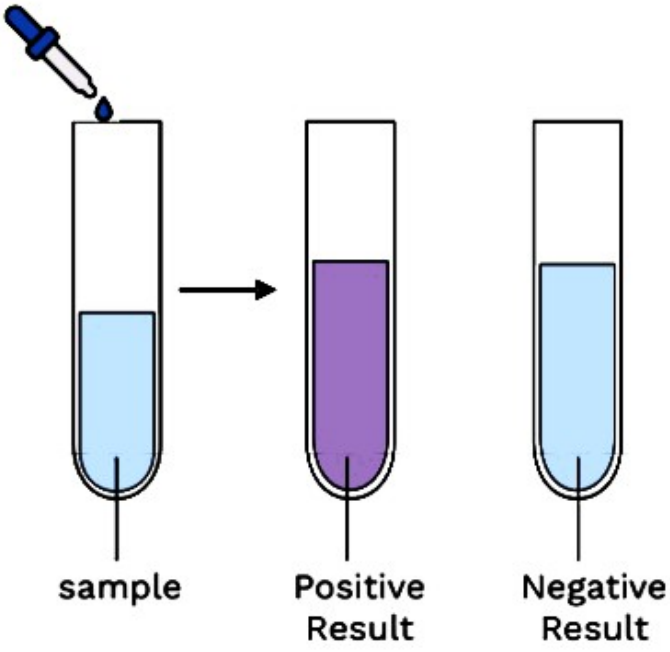
NOTE: Biuret Test are not α-amino acid test. Biuret test is dipeptide test

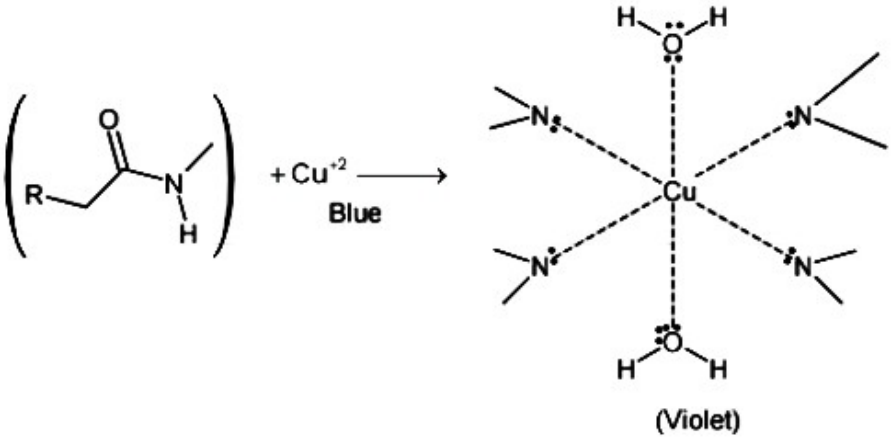
When a protein reacts with copper (II) sulphate (blue), the positive tests is the formation of a violet coloured complex.



Peptide

Peptide-opper complex
(Deep purple)



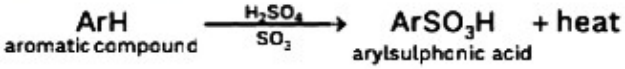


The Biuret test works for any compounds ontaining two one more of the following groups.

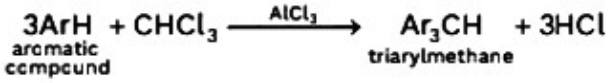
8. Tests for Aromatics

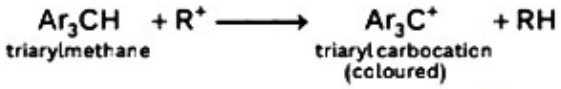
New substituents can be added onto the aromatic ring or existing substituents can be modified, such that the new compound may be more readily characterized.

(i) Fuming Sulphuric Acid



(ii) Chloroform and Aluminium Chloride





Points to remember

- Aromatics give coloured solution or powder.
- Positive Test: (Orange, red, blue, purple, green)
- Non aromatics give yellow colour (Negative result)

Introduction

If the molecule already contains reactive chemical substituents (acids, amines, ethers, carbonyl compounds, etc.,) use the corresponding classification test for that particular group.

Points to remember

- This test is good for aromatics with no other functional groups.
- Positive Test: the aromatic compound dissolves completely in H_2SO_4 with the evolution of heat (Fuming)

(iii) Azoxybenzene and Aluminium Chloride

The colour of the solution or precipitate is dependent on the functional groups present on the aryl group.

Aromatic hydrocarbons and their halogen derivatives produce a deep-orange to dark-red colour in solution or give a precipitate.

Fused aromatic ring (such as naphthalene, anthracene, and phenanthrene) produce brown colour.

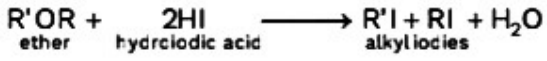
Aliphatic hydrocarbons give no colour or, at most, a pale yellow.

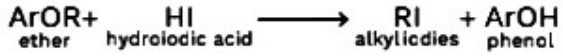
9. Tests for Ethers

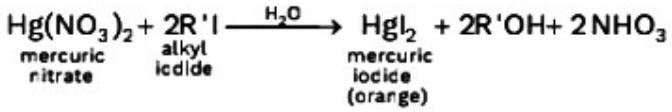
Ethers are only a little more polar and slightly more reactive than either saturated hydrocarbons or alkyl halides.

Ethers form extremely explosive peroxides upon standing, specially when exposed to air and/or light. Liquid ethers that shows solid precipitates should not be handled.

Hydroiodic Acid (Zeisel's, Alkoxyl method)







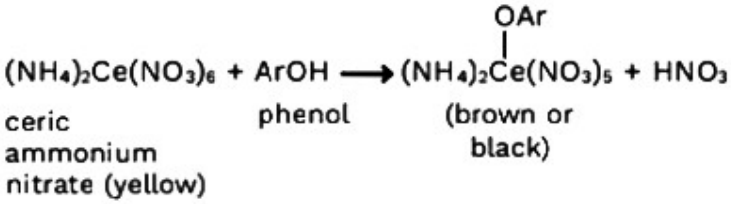
Points to remember

- **Positive Test:** Orange or Orange-red colour indicating.
- **Note :** Ethyl and methyl esters give also positive result.

10. Tests for Phenol

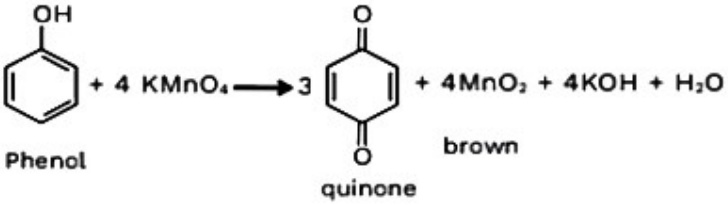
(i) Cerric Ammonium Nitrate Test

Phenols undergo reaction with yellow ceric ammonium nitrate to produce brown or black products.

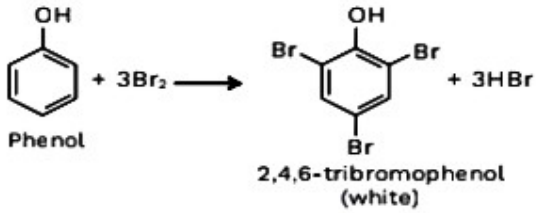


Points to remember

- Phenols reduce potassium permanganate solution and undergo oxidation to quinones.
- The manganese is reduced from +7 which gives a purple solution to +4 which is brown.



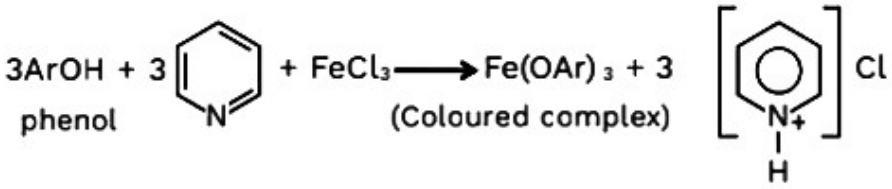
(ii) Bromine Water Test



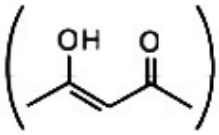
Points to remember

- Positive Test: decolourization of bromine.
- This is good for water soluble phenols.

(iii) Ferric Chloride - Pyridine Reagent



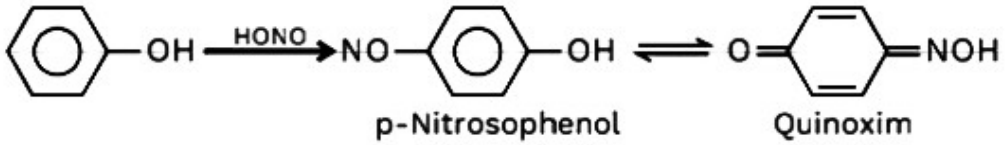
Positive Test: Production of blue, violet, purple, green, or red-brown colours
Good for all types of Ar-OH.

Positive test for enols. 

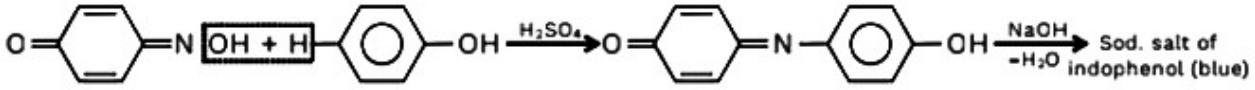
Carboxylic acid with $FeCl_3$ gives red colour only when saturating with NH_3.

(iv) Libermann's nitroso test :

While phenol is reacted with $NaNO_2$ and concentrated H_2SO_4, it provides a deep green colour which changes to red on dilution with water. while generated alkaline along with NaOH, blue colour is restored. This reaction is termed as Liebermann's nitroso reaction and is employed as a test of phenol.



p-Nitrosophenol Quinoxim



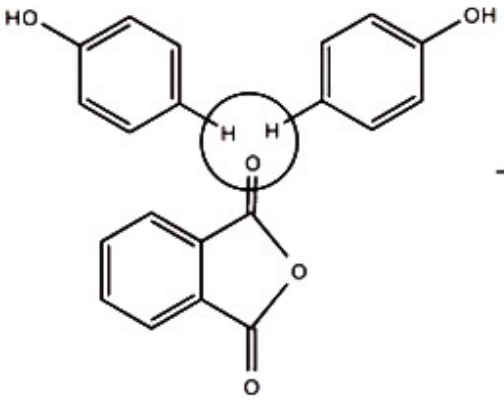
Indo phenol (Red)

(v) With Phthalic anhydride-

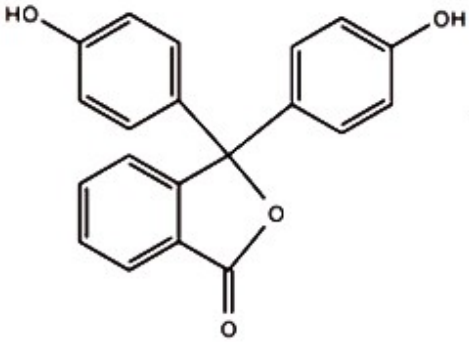
Many Phenols undergo condensation with Phthalic anhydride to give useful product.

Let's understand

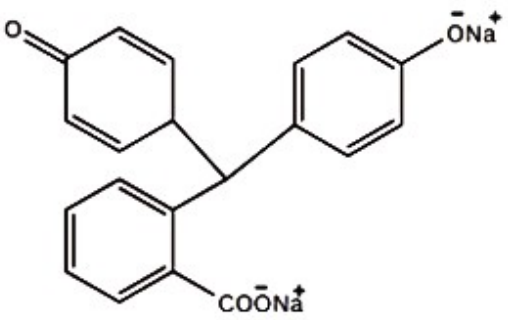
Phenol when heated with phthalic anhydride in the presence of concentrated H_2SO_4 given phenolphthalein, a colourless compound which produce beautiful pink colour with alkali solution due to the formation of coloured sodium salt.



Conc. H_2SO_4



NaOH



Phenolphthalein (Colourless)

Sodium slt of Phenophthalein (Pink)

11. Test For Nitro Compounds

(i) Ferrous Hydroxide Reduction

$$\underset{\text{Nitroalkene}}{RNO_2} + 6\underset{\substack{\text{Ferrous}\\ \text{hydroxide}\\ \text{(green)}}}{Fe(OH)_2} + 4H_2O \longrightarrow \underset{\substack{\text{Ferrichydroxide}\\ \text{(red-brown, brown)}}}{RNH_2 + 6Fe(OH)_3}$$

Note :

Nitroso compounds, quinones, hydroxylamines, alkyl nitrates give also positive results

Points to remember

- Positive Test: indicated by the change in colour from green to red-brown or brown due to the oxidation of iron from+2 to +3.
- A negative test is indicated by a greenish precipitate.

(ii) Zinc and Ammonium Chloride Reduction

(Mulliken–Baker Test)

$$\underset{\text{Nitro compound}}{RNO_2} + 4[H] \xrightarrow[NH_4Cl]{Zn} \underset{\text{Hydroxylamine}}{RNHOH} + H_2O$$

Points to remember

- Test the solution with Tollens Reagent
- **Positive Test:** formation of metallic silver Only 3° aliphatic o compounds and aromatic nitro compounds are reduced by Zn/ NH_4Cl to the hydroxylamine.

$$RNHOH + 2Ag(NH_3)_2OH \xrightarrow[\text{Tollensreagent}]{} RNO + 2H_2O + 2Ag(s) + 4NH_3$$

The hydroxylamine is then detected by the formation of metallic silver in the Tollens test (or formation of a black or grey precipitate). This is known as

QUALITATIVE ANALYSIS OF ORGANIC COMPOUNDS

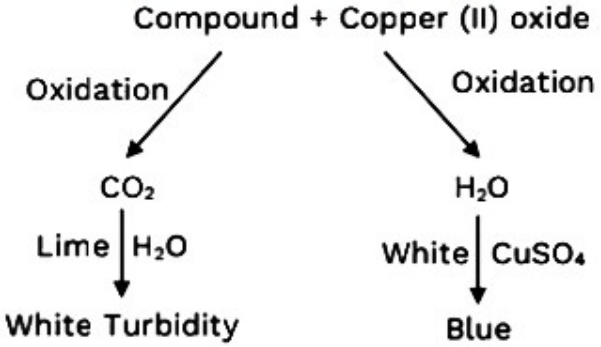
Elements present in organic compounds

(i) Carbon
(ii) Hydrogen
(iii) Oxygen
(iv) Nitrogen
(v) Sulphur & phosphorus

C, H and O are major component in any organic compound, N ,S, P are presenet in less percentage.

Element Detection Test

1. **Detection of Carbon and Hydrogen :**



$$C + 2CuO \xrightarrow{\Delta} 2Cu + CO_2$$

$$2H + CuO \xrightarrow{\Delta} Cu + H_2O$$

$$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 \rightarrow + H_2O$$

$$5H_2O + CuSO_4 \rightarrow CuSO_4 . 5H_2O$$

White Blue

2. Detection of Other Elements

Lassaigne's Test

Organic Compound (N, S, X & P containing)

Fused with sodium metal

Fused Mass

Distillation

Sodium Extract

Following reactions take place:

$Na + + N \longrightarrow NaCN$

$2Na + S \xrightarrow{\Delta\Delta} Na_2S$

$Na + X \xrightarrow{\Delta} NaX$

(X = Cl, Br or I)

C, N, S and X come from organic compound.

(A) Test of Nitrogen

(i) $6CN^- + Fe^{+2} \longrightarrow \underset{(A)}{[Fe(CN)_6]^{-4}}$

(ii) $[Fe(CN)_6]^{-4} + 4Fe^{+3} \xrightarrow{XH_2O} \underset{\text{Prussion Blue}}{Fe_4[Fe(CN)_6]_3 \cdot XH_2O}$

In step (i) Extract react with iron sulphate and form (A).
Further oxidation of (A) with sulphuric, some iron (II) oxidised to iron (III) ions.
Iron (III) react with hexacyanoferrate (II) to produce Iron(III) hexacyanoferrate complex, Prussian blue in colour.

(B) Test for Sulphur

(i) $S.F.E \xrightarrow[CH_3COOH]{H^{\oplus}} \xrightarrow{Pb^{+2}} PbS$ Black ppt

Reaction

(ii) $S.F.E \xrightarrow[\text{Nitroprosside}]{\text{Sodium}}$ Violet Color

Reaction

Special Case

$Na + C + N + S \rightarrow NaSCN$

$Fe^{3+} + SCN^- \rightarrow [Fe(SCN)]^{2+}$

blood red

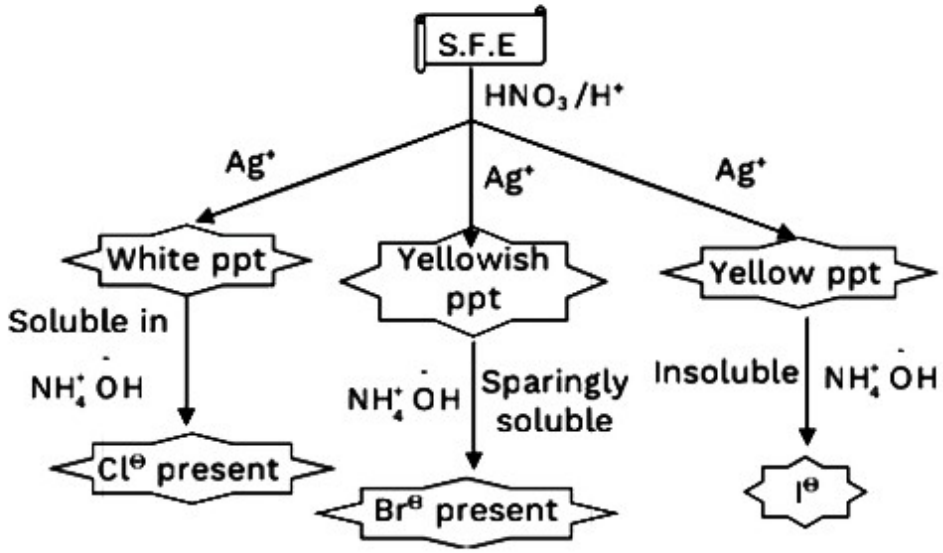
Element Separation Technic

$NaSCN + 2Na \rightarrow NaCN + Na_2S$

Points to remember

In case, nitrogen and sulphur both are present in an organic compound, sodium thiocyanate is formed. It gives blood red colour and no Prussian blue since there are no free cyanide ions.

(C) Test of Halogens

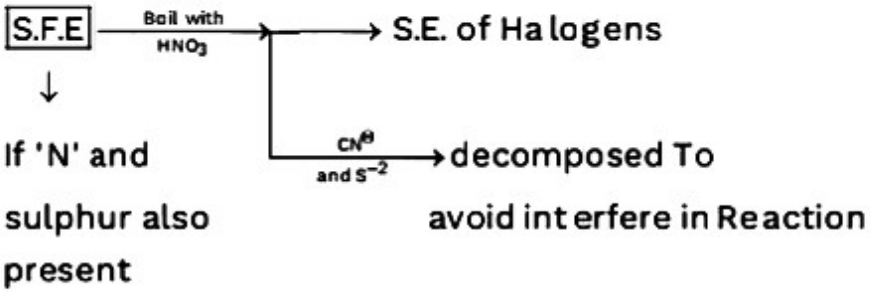


$X^\ominus + Ag^\oplus \longrightarrow AgX$

X represents a halogen Cl, Br, or I

Special Concept

Why boiling required before $AgNO_3$ addition?



Q20 **Lasagne's test for the detection of nitrogen fails in:**

(A) $NH_2CONHNH_2$ **(B)** NH_2NH_2

(C) NH_2CONH_2 **(D)** $C_6H_5NHNH_2$

A20 **B**

For lassaigne's test of nitrogen, compound must contain N along with carbon to from NaCN with sodium extract.

Q21 **The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is-**

(A) $Fe_4[Fe(CN)_6]_3$ **(B)** $Na_3[Fe(CN)]_6$

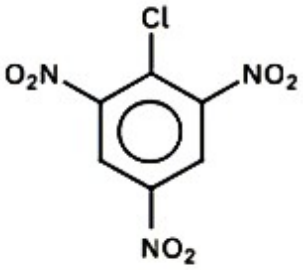
(C) $Fe_3[(CN)_4]SO_4$ **(D)** $Na_4[Fe(CN)_5NOS]$

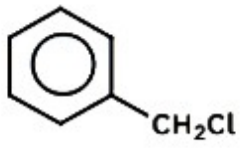
A21 **A**

Q22 **Detection of chlorine is not possible without preparing extract in:**

(A) 

(B) $CH_2=CH-Cl$

(C) 

(D) 

A22 **B**

Vinyl chloride does not react with silver nitrate solution because bond strength is more due to double bond character in C – Cl bond.

(D) Test for Phosphorus

The compound is heated with an oxidising agent (sodium peroxide). The phosphorus present in the compound is oxidised to phophate. The solution is boiled with nitric acid and then treated with ammonium molybdate. A yellow colouration or precipitate indicates the presence of phophorus.

$$Na_3PO_4 + 3HNO_3 \rightarrow H_3PO_4 + 3NaNO_3$$

$$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow (NH_4)_3PO_4 \cdot 12MoO_3 + 21NH_4NO_3 + 12H_2O$$

Ammonium molybdate Ammonium phophomolybdate

2. **Estimation of nitrogen.** Nitrogen is generally estimated by two methods:
(A) Duma's method and (B) Kjeldahl's method.
(A) Duma's method. This method can be applied to all nitrogenous organic compounds.
Principle. The method is based on the fact that when an organic compound is heated with copper oxide, its carbon, hydrogen, sulphur and nitrogen contents are converted

into CO_2, H_2O, SO_2 and N_2 respectively. In case of any traces of nitrogen oxides are formed, these are reduced back to free nitrogen by passing it over bright reduced copper gauze. Carbon dioxide, water vapour and sulphur dioxide are absorbed in caustic potash solution and free nitrogen is collected over KOH in a nitrometer. From the volume of nitrogen collected over caustic potash, percentage of nitrogen can be calculated.

$$C + 2CuO \xrightarrow{heat} CO_2 + 2Cu$$
$$2H + CuO \xrightarrow{heat} H_2O + Cu$$
$$2N + CuO \xrightarrow{heat} N_2 + \text{Oxides of N}$$
$$\text{Oxides N} + Cu \xrightarrow{heat} N_2 + CuO$$

Room temperature and atmospheric pressure are also recorded in order to convert the volume of nitrogen collected under given conditions to volume of N_2 at N.T.P.

$$\therefore \quad \text{Percentage of nitrogen} = \frac{28 \times \text{Vol. of } N_2 \text{ collected at N.T.P} \times 100}{22400 \times \text{Wt. of organic compound}}$$

$$= \frac{\text{Wt. of nitrogen} \times 100}{\text{Wt. of organic compound}}$$

Q23 **0.290 gm of an organic compound containing nitrogen gave 50.00 ml of nitrogen at N.T.P. What is the percentage of nitrogen in the compound?**

A23 Wt. of organic substance = 0.290 gm

Vol. of collected nitrogen at N.T.P. = 50 ml

Now we know that percentage of nitrogen

$$= \frac{28 \times \text{Vol. of } N_2 \text{ collected at N.T.P} \times 100}{22400 \times \text{Wt. of organic compound}}$$

$$= \frac{28 \times 50 \times 100}{22400 \times 0.290} = 21.55$$

Q24 **In the determination of nitrogen by Duma's method, 0.200 gm of compound gave 20.7 ml of moist nitrogen at 15°C and 776 mm pressure. Calculate the percentage of nitrogen in the compound**
(Aq. tension at 15°C = 16 mm ; 1 litre of nitrogen of N.T.P. weighs 1.25 gm) at NTP.

A24 Conversion of volume of N_2 collected to N.T.P. conditions

At given conditions, $P_1 = 776 - 16 = 760$ mm,

$V_1 = 20.7$ ml, $T_1 = 15 + 273 = 288$ K

At N.T.P. conditions, $P_2 = 760$ mm

$V_2 = ?$ $T_2 = 273$ A

Applying the general gas equation

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad \text{or} \quad V_2 = \frac{P_1V_1 \times T_2}{T_1 \times P_2}$$

$$V_2 = \frac{760 \times 20.7 \times 273}{288 \times 760} = 19.62\,\text{ml}$$

Calculation of percentage of nitrogen

Since one litre (1000 ml) of nitrogen at N.T.P weighs = 1.25 gm

$\therefore$ 19.62 ml. of nitrogen at N.T.P. weigh = $\frac{1.25}{1000} \times 19.62\,\text{gm}$

Hence Percentage of nitrogen in the compound = $\frac{1.25}{1000} \times 19.62 \times \frac{100}{0.200}$ = **12.26 %**

(B) Kjeldahl's method. This method is simpler and convenient, and can be used in compounds in which nitrogen is directly linked to carbon and hydrogen. Therefore, this method is largely used for the estimation of nitrogen in food, fertilizers and drugs. However, the method fails in compounds containing nitrogen joined either to other nitrogen or to oxygen as in nitro, nitroso, azo (N = N), azoxy, diazo, hydrazones, oximes, etc. The method is not applicable to compounds containing nitrogen in ring (e.g. pyridine, quinoline etc.)

Principle. This method is based on the fact that most of the nitrogenous compounds when heated with concentrated sulphuric acid in a long-necked flask called Kjedahl's flank in the presence of potassium su1phate and a little copper sulphate **(Kjeldahlisation or digestion)**, its nitrogen is quantitatively converted into ammonium sulphate. The ammonium sulphate so obtained is decomposed with excess of NaOH solution to give ammonia which is absorbed in an excess of a standard HCl or H_2SO_4 solution. The residual mineral acid is now titrated against a standard sodium hydroxide solution and thus the amount of HCl consumed for the neutralisation of ammonia, and hence the amount of ammonia produced from the known weight of the organic compound is determined.

Organic compound containing N + $H_2SO_4 \xrightarrow{\text{heat}} (NH_4)_2SO_4$

$(NH_4)_2SO_4 + 2NaOH \xrightarrow{\text{heat}} Na_2SO_4 + 2NH_3 + 2H_2O$

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$

Hence percentage of nitrogen in the compound

$$= \frac{14 \times \text{Vol. of acid used} \times \text{Normality of acid}}{1000 \times \text{Wt. of organic compound}} \times 100$$

Type I: *When the ammonia evolved completely neutralises the standard acid taken.*

Q25 **If the ammonia obtained from 0.2160 gm of an organic compound by Kjeldahl method of analysis neuralises 23.85 ml of 0.1150 N acid, what is percentage of nitrogen in the compound?**

A25 Wt. of substance taken = 0.2160 gm
Vol. of acid neutralised by NH_3 = 23.85 ml
Normality of the acid = 0.1150 N

$$\therefore \text{Percentage of nitrogen} = \frac{1.4 \times N \times V}{\text{Wt. of substance}} = \frac{1.4 \times 0.1150 \times 23.85}{0.2160} \text{ gm} = 17.77$$

Type II. When the ammonia evolved neutralises part of the standard acid and the acid left unused is titrated against standard NaOH.

Q26 **0.445 gm of an organic compound was treated according to Kjeldahl method. The ammonia evolved was absorbed in 100 ml of N/2 sulphuric acid. The excess of the acid required 65.5 ml. N/2 alkali solution for neutralisation. Find the percentage of nitrogen in the compound.**

A26 Wt. of organic compound = 0.445 nm
Vol. of N/2 H_2SO_4 taken = 100 ml.
Vol. of N/2 alkali required for the neutralisation of the excess of the acid = 65.6 ml.
$\therefore$ Vol. of N/2 acid used for neutralising NH_3 = 100 – 65.5 ml = 34.5 ml.
Hence, percentage of nitrogen in the compound

$$= \frac{1.4 \times N \times V}{\text{Wt. of substance}} = \frac{1.4 \times 1 \times 34.5}{2 \times 0.445} = 54.33\%$$

Type III. *When the residual acid in the conical flask is diluted to a known volume with water and then titrated against standard alkali solution.*

Q27 **0.6 gm of an organic compound was Kjeldahlised and the ammonia evolved was passed into 100 ml of seminormal H_2SO_4. The residual acid was diluted to 500 ml with distilled water; 25 ml. of diluted acid required 20.4 ml decinormal caustic soda solution for complete neutralisation. Calculate the percentage of nitrogen in the compound.**

A27 Weight of substance taken = 0.6 gm

Vol. of N/2 H_2SO_4 taken = 100 ml

Let the vol. of N/2 H_2SO_4 unused = V_1 ml

Now 25 ml of the diluted acid solution = 20.4 ml N/10 NaOH

$\therefore$ Normality of the diluted acid = $20.4 \times \frac{1}{10} \times \frac{1}{25} = \frac{51}{625}N$

500 ml. of $\frac{51}{625}N$ of diluted acid $\equiv$ V_1 ml of N/2 H_2SO_4 (unused)

Applying the normality formula

Eq. (for unsed acid) = Eq. (for diliuted acid)

$$\frac{N}{2} \times V_1 = \frac{51}{625}N \times 500$$

$\therefore$ $$V_1 = \frac{51}{625} \times 500 \times \frac{2}{1} = 81.6 \text{ ml}$$

$\therefore$ Vol. of N/2 acid used for ammonia = 100 – 81.6 ml = 18.4 ml.

Hence percentage of nitrogen = $\frac{1.4 \times N \times V}{w} = \frac{1.4 \times 1 \times 18.4}{2 \times 0.6} = \mathbf{21.46}$

3. **Estimation of halogens (Carius method). Principle.** Carius method is based on the fact that when an organic compound containing halogen is heated in a sealed tube with fuming nitric acid in presence of silver nitrate; C, H and S of the compound are oxidised to their respective volatile oxides, while halogen is converted to the silver halide. From the weight of the silver halide formed, percentage of halogen in the organic compound can be calculated.

$$\text{In general, \% of halogen} = \frac{\text{At. wt. of halogen} \times \text{Wt. of AgX} \times 100}{\text{Mol. wt. of AgX} \times \text{w}}$$

Note

(i) The method does not give satisfactory results with iodine because silver iodide is slightly soluble in nitric acid and some molecular iodine is also produced even in presence of excess of silver nitrate.

(ii) The method does not give correct result for highly halogenated aromatic compounds.

4. **Estimation of sulphur** (Carius method). Sulphur is estimated like halogens but with a slight modification [here $AgNO_3$ crystals used in halogen estimation are not used). A weighed amount of organic substance (let it be w gm) is heated with cone. HNO_3 when its C, H and S are oxidised CO_2, H_2O (steam) and H_2SO_4 respectively. The contents of the

Carius tube are transferred to a beaker and then treated with excess of barium chloride solution when whole of the sulphuric acid is precipitated as barium sulphate. From the weight of $BaSO_4$ (say x gm), percentage of S can be calculated.

$$C + 2O(\text{from } HNO_3) \xrightarrow{\text{heat}} CO_2;\ 2H + O(\text{from } HNO_3) \xrightarrow{\text{heat}} H_2O$$

$$S + H_2O + 3O(\text{from } HNO_3) \xrightarrow{\text{heat}} H_2SO_4$$

$$H_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2HCl$$

$$\therefore \text{ Percentage of S in the compound} = \frac{32 \times x \times 100}{233 \times w}$$

Q28 **0.2595 gm of an organic compound containing sulphur was heated in a Carius tube with fuming nitric acid and the sulphuric acid formed was precipitated as $BaSO_4$ with $BaCl_2$. The weight of dry $BaSO_4$ was 0.350. Find the percentage of sulphur in the substance.**

A28 $$\% \text{ of sulphur} = \frac{32}{233} \times \frac{\text{Wt. of } BaSO_4 \times 100}{\text{Wt. of compound}} = \frac{32}{233} \times \frac{0.350 \times 100}{0.2595} = \mathbf{18.52}$$

5. **Estimation of oxygen.** There is no satisfactory method for the direct estimation of oxygen in an organic compound. Hence, it is usually calculated by difference. In case, the sum of percentages of all other elements present in the organic compound is less than 100, the remainder gives the percentage of oxygen, i.e.,
% of oxygen = 100 – Sum of percentages of all other elements

(A) Victor Meyer's method. Mol. wt = $\frac{\text{Weight of compound} \times RT}{PV}$

Where V is the volume in litres collected at pressure P atmosphere and temperature T (in Kelvin) from w g of solute, R = 0.082 L atom mol^{-1} K^{-1}.

3. **Calculation of molecular formula.** The formula which represents the actual number of atoms of various elements present in one molecule of the substance is termed as molecular formula. It is either the same as the empirical formula of the substance or a

simple multiple of it, i.e.

Molecular formula = $(\text{Empirical formula})_n$

Where n is a whole number and is the ratio between the molecular weight and the empirical formula weight of the substance, i.e.

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$

Q29 **An suspension of $CaOCl_2$ in water is heated with ethanol. The product formed is:**

(A) Ethylene **(B) Ethanal**

(C) Trichloromethane **(D) Chloroethane**

A29 **C**

$CH_3 - CH_2 - OH + CaOCl_2 \rightarrow CHCl_3 + (HCOO)_3Ca$

Q30 **A product of yeast fermentation of sugars in grains and fruits is :**

(A) Ethanol **(B) Isobutanol**

(C) Isopropanol **(D) Methanol**

A30 **A**

Ethanol is produced by the fermentation of sugar in grain & fruits. It is also known as grain alcohol.

Q31 **Which of the following compound does not dissolve in conc. H_2SO_4 even on warning?**

(A) Ethylene **(B) Benzene**

(C) Hexane **(D) Aniline**

A31 **C**

Ethylene absorbs H_2SO_4 forming $CH_3 - CH_2OSO_3H$ and dissolve.

Benzene with warm H_2SO_4 undergoes sulphonation, and dissolve.

Aniline with H_2SO_4 forms anilinium hydrogen sulphate salt and dissolve.

Hexane, a hydrophobic molecule, does not react with H_2SO_4, remains insoluble.

Q32 **Which of the following phenols is most soluble in aqueous sodium bicarbonate?**

(A) 2,4-Dihydroxyacetopheone **(B) p-Cyanophenol**

(C) 3,4-Dicyanophenol **(D) 2,4,6-Tricyanophenol**

A32 **D**

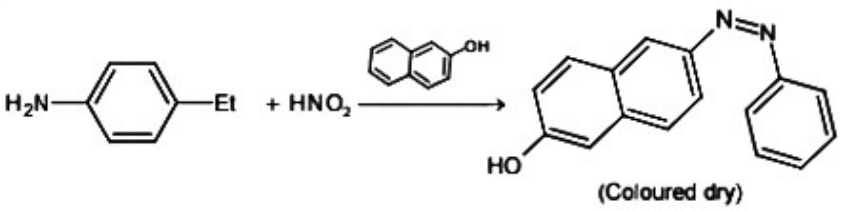
2,4,6- Tricyanophenol is stronger acid than sodium bicarbonate, so it can give CO_2 gas with $NaHCO_3$

Q33 **Compound (Z) on reaction with nitrous acid followed by reaction with β-naphthol in alkaline solution gives bright colour dye. Compound Z will be:**

(A) N,N-Dimethylaniline **(B) N-Ethyl-N-methylaniline**

(C) Benzylamine **(D) Para-Ethyl aniline**

A33 **D**



Q34 **Which of the following reagents may be used to distinguish between phenol and picric acid?**

(A) Aqueous NaOH **(B) Tollen's reagent**

(C) Molisch reagent **(D) Neutral $FeCl_3$**

A34 **D**

Q35 **Which of the following reagents may be used to distinguish between carbohydrate and aldehydes?**

(A) Aqueous NaOH **(B) Tollen's reagent**

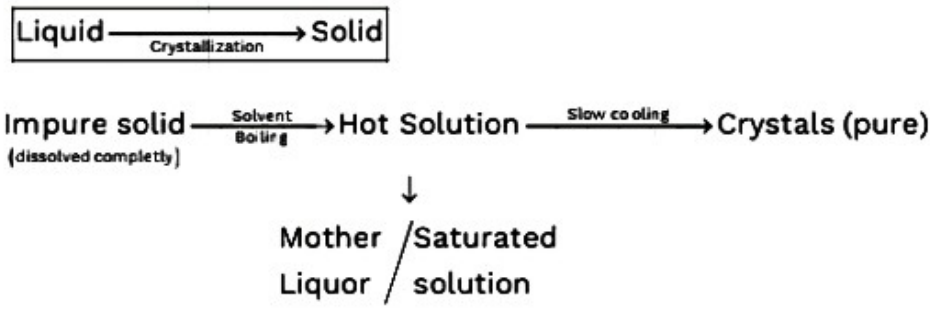
(C) Molisch reagent **(D) Neutral $FeCl_3$**

A35 C

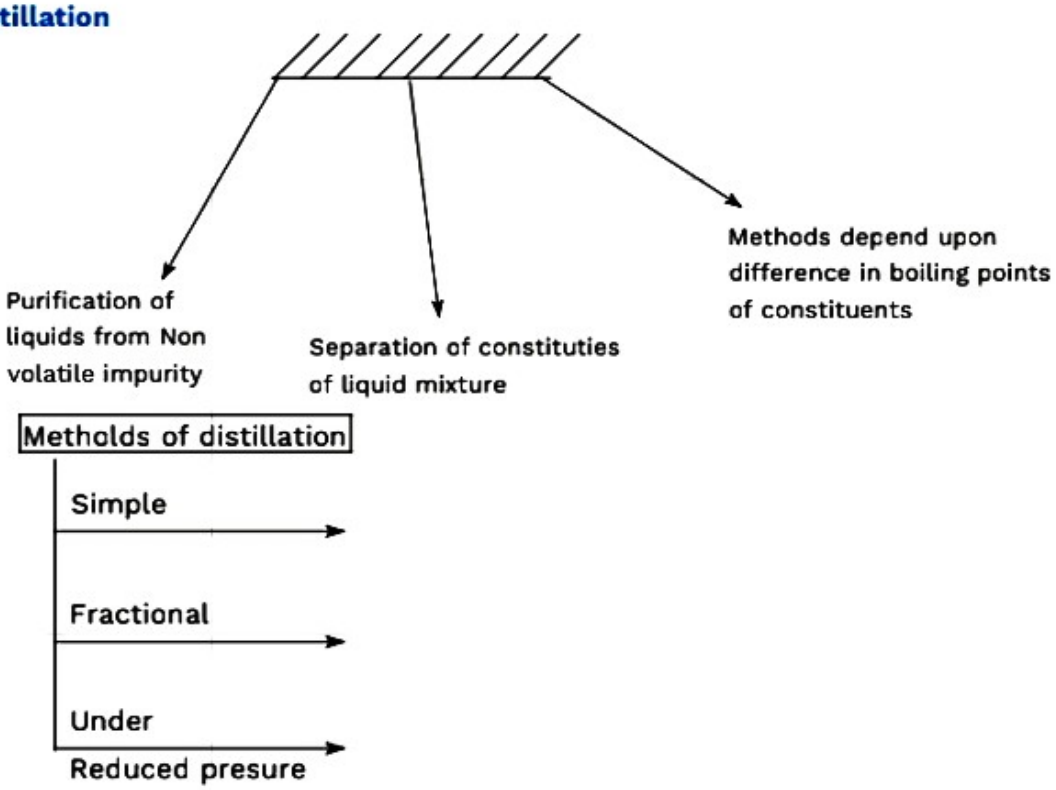
Sublimation

Applicable which turn into vapour directly from solid if heated.
Benzoic Acid, Naphthalene and Camphor are some known examples.
Separation techniques for volatile and non-volatile solids

Crystallization



Distillation

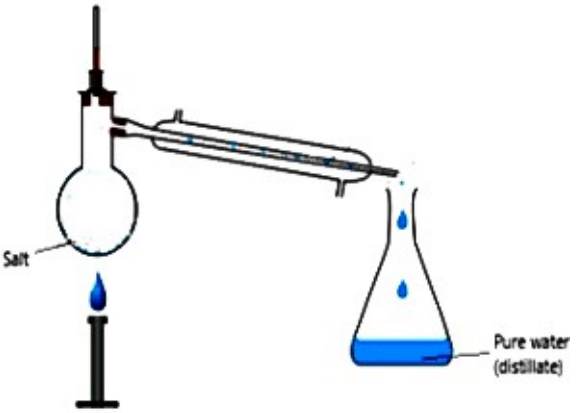


Simple Distillation

Condition:

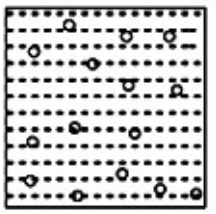
(i) Large difference in B.P.

(ii) Do not decompose under ordinary pressure



Fractional distillation

Fractional Distillation

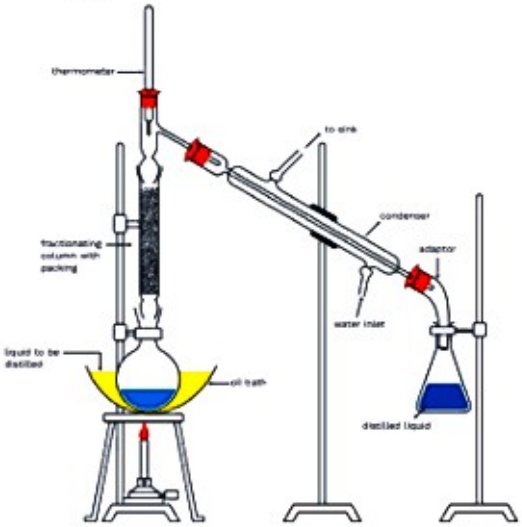


Liquid 'A'

Liquid 'B'

Liquid A = '–'
Liquid B = 'O'

* Low difference in boiling point



Example

Petroleum Coal-tar and Crude Oil

Steam distillation:

Condition for steam distillation of compound: Note: Applicable for solid & liquids both

- Do not decompose at the steam temp.
- Should have fairly high vapour pressure.
- Insoluble in water.
- Non-volatile impurities.

Uses :

(i) Recovery of essential oils from plants and flowers.

(ii) Manufacture of aniline and turpentine oil.

Distillation Under Reduced Pressure

Azeotropic Distillation

- Useful for mixture of liquids that can't separated by fractional distillation.
- What is called azeotropes?
- which can be purified only by azeotropic distillation.

Concept behind azeotropic distillation :

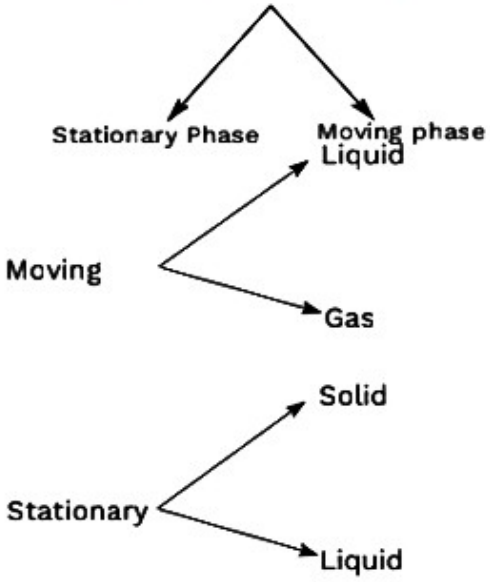
- Add third component like benzene CCl_4, (-0-), glycerol, glycol.
- Third component act like dehydrating agent supress the partial pressure of one component.
- Raise B.P. of other component so distillation will give one component will distil over.

Differential Extraction :

- Removing a component from its aqueous solution by shaking with appropriate organic solvent.

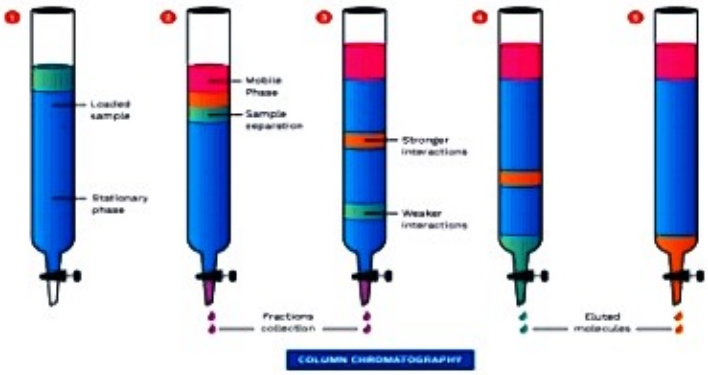
Chromatography:

- Concept behind chromatography :
  Selective distribution of the mixture of organic substances between two phase.



Chromatography :

- Column chromatography (CC)
- Thin layer (T.L.C)
- Paper chromatography (P.C)



Adsorption chromatography :

- Adsorbent = Silica gel and alumina
- Principle = different degree of adsorption on an adsorbent.
- Based on principle of different adsorption.
- Movement of mixture by varying distances over the stationary phase.

Partition chromatography

Example : Paper chromatography

- Strip of paper acts as an adsorbent
- Similar to thin layer chromatography.
- Paper strip act as stationary phase.
- Paper is known chromatography paper.
- Differential method of portioning of components of mixture between stationary and mobile phase.

